Exhaustive Exploratory Data Analysis: Accounts Portfolio

Portfolio Risk, Recovery Geometry & Behavioral Underwriting

N = 2,400 Accounts | Total Demand: ₹1.80 Cr/mo

1. LOAN PORTFOLIO COMPOSITION: VOLUME VS. CAPITAL EXPOSURE

Graph 1.1: Monthly EMI Demand by Product (Total Demand: ₹1.80 Crore)

Capital Concentration



- PL Salaried: ₹68.84L (38.2%)
- MSME: ₹58.89L (32.7%)
- Credit Card: ₹21.08L (11.7%)
- Two-Wheeler: ₹17.57L (9.8%)
- Consumer Durable: ₹8.22L (4.6%)
- MFI JLG: ₹5.42L (3.0%)

The loan book demonstrates extreme capital concentration: **MSME and Salaried PLs account for 70.9% of total monthly EMI cash flow** (₹1.28 Cr/mo) despite comprising only 38.5% of debtor volume. Conversely, Microfinance (MFI) and Consumer Durable accounts represent high customer headcounts but minimal monthly capital footprint (₹1,513 average EMI).

2. DELINQUENCY AGING FUNNEL & DISCRETE INSTALLMENT ESCALATION

Graph 1.2: Aging Funnel Progression & Integer Multiplier Dynamics

Delinquency Cascade

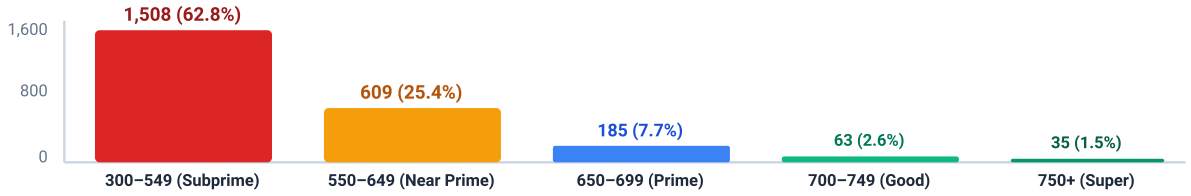
Bucket X (Current)	Bucket 1–30	Bucket 31–60	Bucket 61–90	Bucket 90+ (NPA)
361 (15.0%)	664 (27.7%)	528 (22.0%)	377 (15.7%)	470 (19.6%)
0 DPD | 1.0× EMI	15.1 DPD | 1.0× EMI	45.9 DPD | 2.0× EMI	74.8 DPD | 3.0× EMI	205.9 DPD | 6.96× EMI

Deterministic Rule: Overdue Balance is an exact integer multiple of monthly installment: Overdue = n × EMI

Intervention Window: Overdue amounts scale predictably: 1.0× in Bucket 1–30, jumping strictly to 2.0× in 31–60, 3.0× in 61–90, and 6.96× in 90+ NPA. The early cohort (≤30 DPD) holds 42.7% of accounts, representing the optimal ROI window before default becomes systemic.

3. CREDIT BUREAU SCORE SEGMENTATION & SUBPRIME SATURATION

Graph 2.1: Credit Bureau Score Distribution (88.2% Impaired Base) Credit Risk Profile



Nearly two-thirds of borrowers (**62.8%, 1,508 accounts**) reside in the deep subprime bracket (<550). Across the whole book, **88.2% possess credit scores below 650**. Because subprime debtors are habituated to damaged credit scores, legal warnings and bureau reporting threats have minimal leverage; recovery requires UPI convenience and restructuring.

4. BORROWER INCOME ARCHETYPES & FINANCIAL VOLATILITY

Graph 2.2: Income Source Breakdown & Cash Flow Commitment Cash-Flow Archetypes

Salaried (53.5%)
1,283 Accounts
Mean EMI: ₹7,292
Capped at ₹18,000 | Payday lag

Self-Employed (33.0%)
792 Accounts
Mean EMI: ₹9,989
Peak ₹39,950 | Trade cycle variance

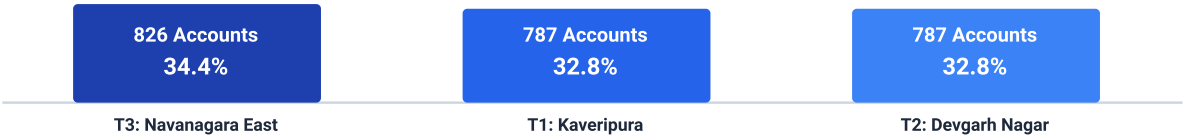
Daily Wage (13.5%)
325 Accounts
Mean EMI: ₹2,255
Micro-repayment | Highly fragile

Income Archetype	Account Volume	Average Monthly EMI	Max Single EMI	Optimal Outreach Cadence
Salaried	**1,283 (53.5%)**	**₹7,292**	**₹18,000**	+24h post salary credit date (Days 1–10)
Self-Employed	**792 (33.0%)**	**₹9,989**	**₹39,950**	Post-weekend settlement / Monday trade cycle
Daily Wage	**325 (13.5%)**	**₹2,255**	**₹4,800**	Split daily/weekly micro-settlement installments

5. GEOGRAPHIC MARKET CENTERS: HOMOGENOUS TOWN FOOTPRINT

Graph 3.1: Geographic Account Allocation (2,400 Total Accounts)

Territory Dispersion



Debtor accounts are split across operating hubs in a near-perfect 1:1:1 ratio. **Navanagara East (34.4%)**, **Kaveripura (32.8%)**, and **Devgarh Nagar (32.8%)** show zero localized geographic concentration risk. Field collections, agency allocations, and telephony caller pools can thus be staffed symmetrically.

6. LINGUISTIC PROFILING & TELEPHONY ROUTING CALIBRATION

Graph 3.2: Customer Language Preference Breakdown

Acoustic Contact Matching



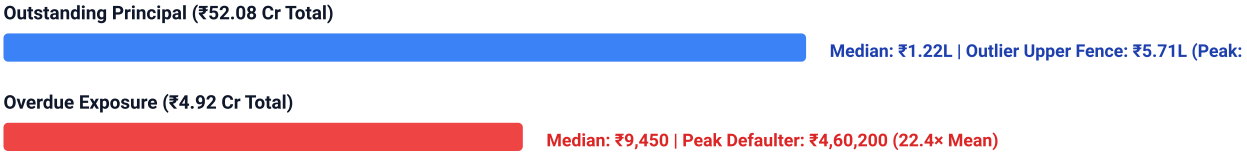
Hinglish (1,217 Accounts): Primary conversational dialect for T2 (Devgarh) and T3 (Navanagara).
Kanglish (640 Accounts): Essential regional dialect for Kaveripura (T1) collection agents.
English (543 Accounts): Corporate salaried professionals concentrated in Navanagara East.

Language Preference	Total Accounts	Primary Territory	Collection Telephony Routing Strategy
Hinglish	**1,217 (50.7%)**	T2 Devgarh & T3 Metro	Conversational Hindi/Hinglish IVR voice bots
Kanglish	**640 (26.7%)**	T1 Kaveripura	Native Kannada tele-callers and vernacular field teams
English	**543 (22.6%)**	T3 Navanagara East	Automated WhatsApp payment links & email notices

7. FINANCIAL EXPOSURE: EXTREME KURTOSIS & OUTLIER DISTRIBUTION

Graph 4.1: Capital Exposure Skewness (Outstanding vs. Overdue)

Kurtosis: +41.47

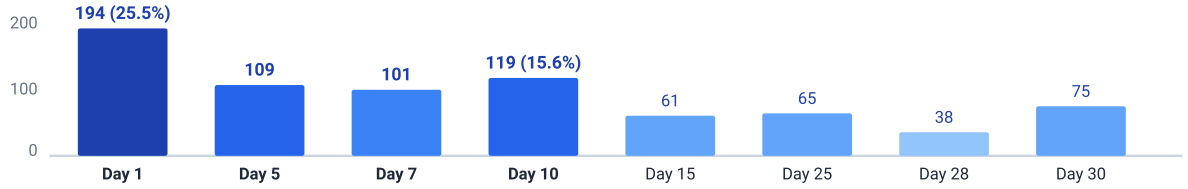


Both `outstanding` and `overdue_start` exhibit severe positive skewness (**+5.31**) and excess kurtosis (**+41.47**). A small cohort of 233 chronic MSME defaults (9.7%) carry overdue amounts exceeding ₹47,225, driving disproportionate portfolio provisions. Predictive machine learning models require log transformations (log(1+x)) to prevent extreme gradient swings.

8. PAYROLL SYNCHRONIZATION & CASH-FLOW TIMING FRICTION

Graph 4.2: Salary Credit Day Spikes (762 Documented Salaried Borrowers)

Mandate Presentation Timing



Mandate Re-Alignment: Over 68.6% of salaried paydays fall between Days 1 and 10 (Day 1 mode = 25.5%). When NACH mandate debits are scheduled mid-month (Day 15–20), borrowers run out of liquidity. Shifting debit presentation to within 48 hours of payday eliminates structural default bounces.

9. NACH MANDATE BOUNCE DIAGNOSTICS: FAILURE MODES

Graph 5.1: Mandate Presentation Failure Root Causes

Technical vs. Behavioral



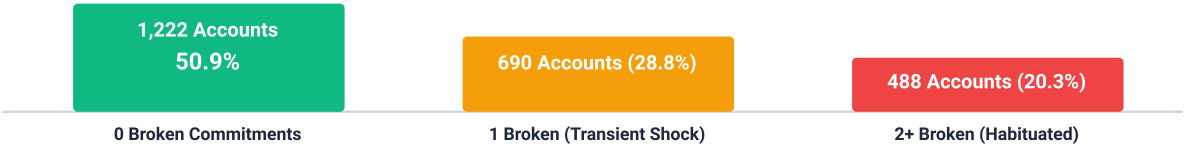
Insufficient Funds (84.1%): Pure cash-flow timing friction; requires pre-debit notifications.
Technical Glitch (6.6%) & Unregistered Mandate (4.4%): 11.0% operational errors fixable via re-clearing.
Stop Payment (4.9%): 118 accounts displaying willful legal dispute or refusal to service.

Diagnostic codes prove that default is predominantly a liquidity friction issue rather than customer fraud. **84.1% of failures are insufficient funds**, which respond well to payday-aligned representation. Only 4.9% reflect active willful intervention (`stop_payment`), while **11.0% are technical/setup failures** requiring no field collector dispatch.

10. PROMISE-TO-PAY (PTP) COMMITMENT INTEGRITY & DECAY

Graph 5.2: Broken PTP Distribution & Behavioral Habituation

Commitment Fatigue



Broken PTP Tier	Account Count	Portfolio Share	PTP Fulfill Probability	Recommended Action
Clean (0 Broken)	1,222	50.9%	> 78%	Standard soft payment link SMS
First Break (1 Broken)	690	28.8%	~ 42%	Follow-up reminder call + UPI link
Chronic (2+ Broken)	488	20.3%	< 14%	Immediate field visit / Doorstep notice

11. MULTI-LENDER EXPOSURE & SELECTIVE DEBT SERVICING

Graph 6.1: Concurrent Lenders & 30-Day Cross-Institution Payments

Repayment Prioritization

Cross-Institutional Exposure

84.2% of Borrowers

2,021 Borrowers hold 1 to 8 other active credit lines

Average: 1.9 concurrent active loans

Selective Prioritization (30 Days)

606 Accounts (25.3%)

Serviced other lenders while leaving this account delinquent

High collection leverage for phone agents

More than 84% of borrowers maintain concurrent loans elsewhere. Crucially, **25.3% (606 borrowers) paid competing lenders within the last 30 days** while remaining in default with this institution. This demonstrates that one-quarter of default cases do not stem from total insolvency, but from selective prioritization of other facilities.

12. DIALLING STRATEGY ARM: RANDOMIZED CONTROLLED HOLDOUT

Graph 6.2: Outreach Strategy Allocation (RCT Calibration)

A/B Testing Framework



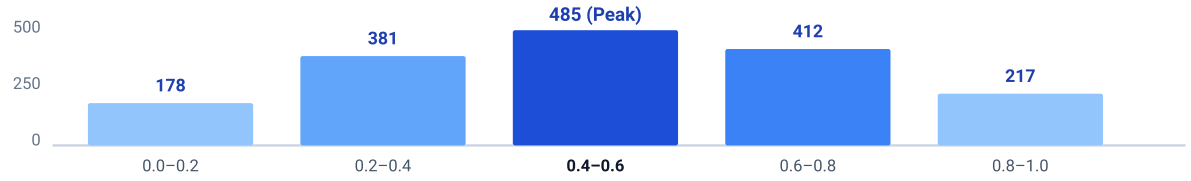
Rule-Based Strategy (94.9%): Calling priority dictated by DPD risk and high-value balance algorithms.

Random Holdout (5.1%, 123 Accts): Unbiased exploration group used to benchmark organic contact yield.

Strategy Arm	Account Volume	Share (%)	Data Science & Modeling Guideline
Rule-Based	**2,277**	**94.88%**	Standard portfolio production policy
Random Contact Point	**123**	**5.12%**	Treat as experimental control covariate, not customer behavior

13. ABILITY-TO-PAY METRIC & COLD-START IMPUTATION ARCHITECTURE

Graph 7.1: Repayment Propensity Score Histogram (1,673 Scored) — Gaussian Bell Curve



The normalized `ability_to_pay_estimate` forms a symmetrical distribution (mean 0.53, std 0.25). Exactly **727 accounts (30.3%) have missing values** due to cold-start telemetry gaps. Missingness is distributed uniformly across bureau score tiers (MAR), allowing gradient-boosted regression to impute missing scores reliably.

14. INTER-FEATURE CORRELATION ARCHITECTURE & MULTICOLLINEARITY

Graph 7.2: Feature Correlation Clusters (Pearson r Network) — Feature Engineering

Capital Collinear Cluster	Commitment Fatigue Cluster	Orthogonal Risk Factor
emi_amount ↔ outstanding	prev_ptp_count ↔ broken	other_active_loans ↔ balance
r = +0.860	r = +0.801	r = +0.001 (Pure Signal)
Engineered: emi / outstanding	Engineered: broken / (count + 1)	Zero collinearity; independent

Strategic Summary & Implementation: High collinearity between installment and balance (r = +0.860) requires ratio transformation (`emi / outstanding`). Commitment fatigue (r = +0.801) demands that repeat breakers bypass verbal promises in favor of same-day UPI links, while independent signals like `other_active_loans` (r = +0.001) offer pure orthogonal risk prediction.